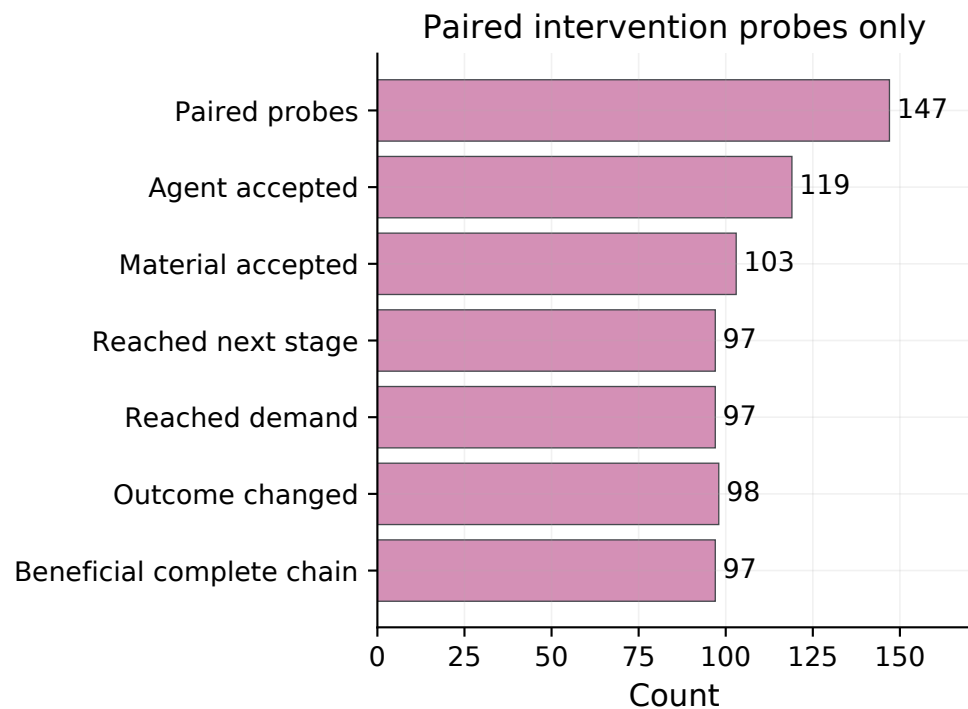
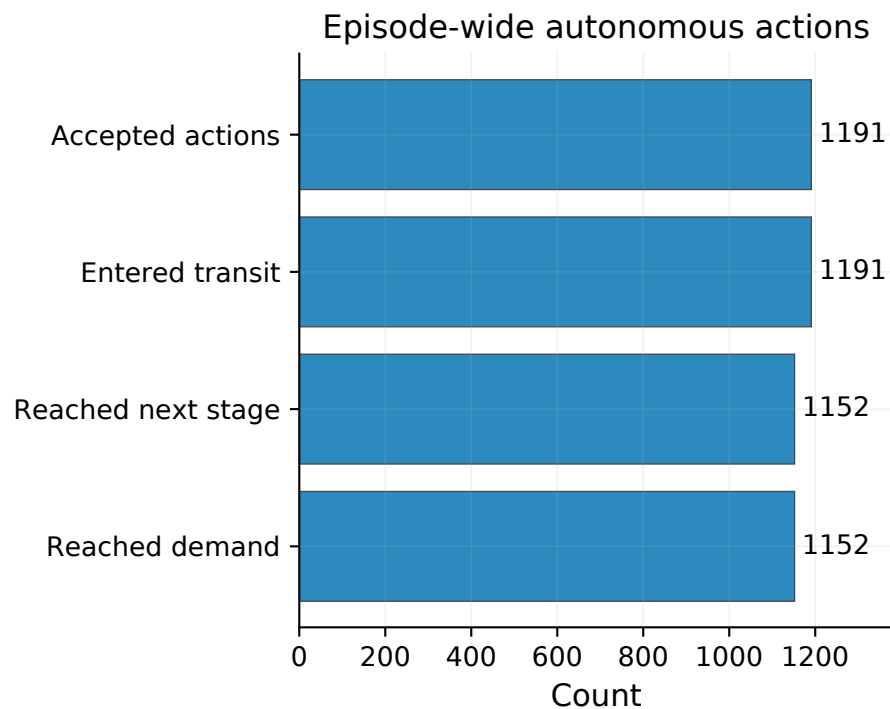


Separate actionability and causal-chain populations — development only



Distinct denominators: left = all autonomous material actions; right = 147 counterfactual probes. Episode context: 220 requests and 119 interventions.